STERLING
INFORMATION TECHNOLOGIES INC

University of Windsor
School of Computer Science

Layered Defense

Lab #6

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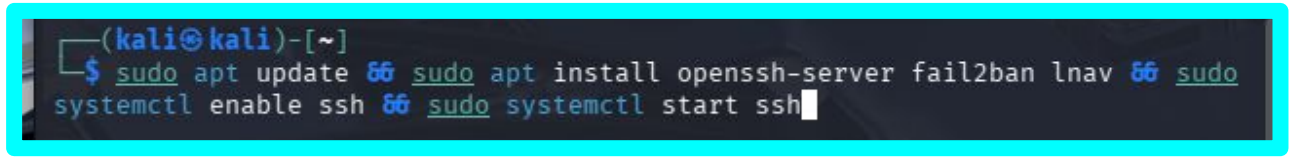
Foreword

- A SIEM(Security Information and Event Management) is used to collect logs from hosts, firewalls and services
- We will be simulating SIEM logs with lnav
- Automation enables response actions like blocking, and alerting
- This lab automates detection and response using a tool called Fail2Ban

Lab Setup

- You will need two VMs one attacker and one defender
- Make sure both are set to host only networks
- Defender setup:

```
sudo apt update
sudo apt install openssh-server fail2ban lnav
sudo systemctl enable ssh
sudo systemctl start ssh
```



Lab Setup

- Check ip address of defender: ip a

```
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 00:0c:29:cd:47:03 brd ff:ff:ff:ff:ff:ff
    inet 172.16.240.129/24 brd 172.16.240.255 scope global dynamic noprefixro
ute eth0
       valid_lft 1381sec preferred_lft 1381sec
    inet6 fe80::759c:d63d:81f0:b165/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

- Stop Fail2Ban for baseline logs:

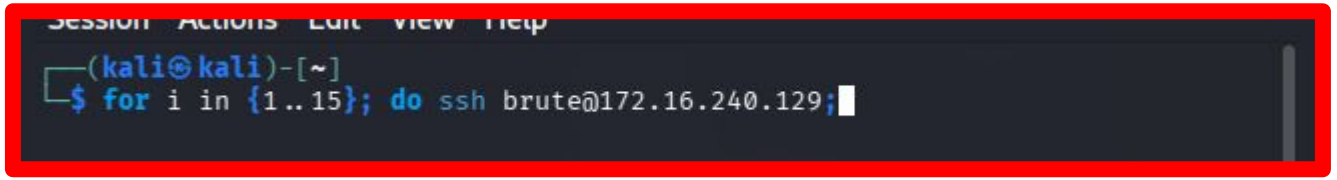
sudo systemctl stop fail2ban

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl stop fail2ban
```

Brute Force Attack Baseline

- On your attacker machine try repeated ssh connections:

```
for i in {1..15};do ssh testuser@172.16.240.129; done
```



- On your defending machine check logs:

```
sudo journalctl -u ssh -o short-iso -f | lnav
```

Baseline Logs

- Observe and screenshot logs to see multiple failed attempts

```
2026-02-21T09:26:04-05:00 kali sshd[5812]: srclimit_penalise: 172.16.240.128/32: activating ipv4 penalty of 18 seconds for penalty: failed authentication
2026-02-21T09:26:04-05:00 kali sshd[5812]: drop connection #0 from [172.16.240.128]:43032 on [172.16.240.129]:22 penalty: failed authentication
2026-02-21T09:26:04-05:00 kali sshd[5812]: drop connection #0 from [172.16.240.128]:43038 on [172.16.240.129]:22 penalty: failed authentication
2026-02-21T09:26:04-05:00 kali sshd[5812]: drop connection #0 from [172.16.240.128]:43048 on [172.16.240.129]:22 penalty: failed authentication
2026-02-21T09:26:04-05:00 kali sshd[5812]: drop connection #0 from [172.16.240.128]:43058 on [172.16.240.129]:22 penalty: failed authentication
2026-02-21T09:26:04-05:00 kali sshd[5812]: drop connection #0 from [172.16.240.128]:43062 on [172.16.240.129]:22 penalty: failed authentication
2026-02-21T09:26:04-05:00 kali sshd[5812]: drop connection #0 from [172.16.240.128]:43068 on [172.16.240.129]:22 penalty: failed authentication
2026-02-21T09:26:04-05:00 kali sshd[5812]: drop connection #0 from [172.16.240.128]:43074 on [172.16.240.129]:22 penalty: failed authentication
```

Enable Automated Defense Layer

- Start Fail2Ban: `sudo systemctl start fail2ban`
- Ensure Fail2Ban is active: `sudo fail2ban-client status sshd`

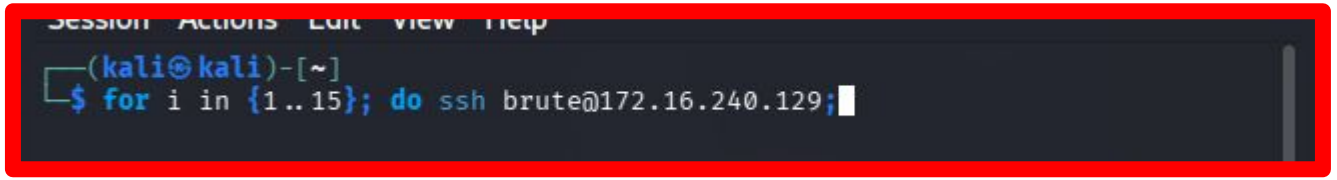
```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl start fail2ban

┌──(kali㉿kali)-[~]
└─$ sudo fail2ban-client status sshd
Status for the jail: sshd
├- Filter
|  ├- Currently failed: 0
|  ├- Total failed:     20
|  `- Journal matches:  _SYSTEMD_UNIT=ssh.service + _COMM=sshd
```

Repeat Brute Force Attack

- On your attacker machine try repeated ssh connections:

```
for i in {1..15};do ssh testuser@172.16.240.129; done
```



- On your defending machine check logs:

```
sudo journalctl -u ssh -o short-iso -f | lnav
```

Fail2Ban Logs

- Observe and screenshot logs

```
2026-02-21T09:28:20-05:00 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
2026-02-21T09:38:10-05:00 kali sshd-session[7215]: Invalid user brute from 172.16.149.128 port 42300
2026-02-21T09:38:14-05:00 kali sshd-session[7215]: pam_unix(sshd:auth): check pass; user unknown
2026-02-21T09:38:14-05:00 kali sshd-session[7215]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=172.16.149.128
2026-02-21T09:38:16-05:00 kali sshd-session[7215]: Failed password for invalid user brute from 172.16.149.128 port 42300 ssh2
2026-02-21T09:38:17-05:00 kali sshd-session[7215]: pam_unix(sshd:auth): check pass; user unknown
2026-02-21T09:38:20-05:00 kali sshd-session[7215]: Failed password for invalid user brute from 172.16.149.128 port 42300 ssh2
2026-02-21T09:38:20-05:00 kali sshd-session[7215]: pam_unix(sshd:auth): check pass; user unknown
2026-02-21T09:38:22-05:00 kali sshd-session[7215]: Failed password for invalid user brute from 172.16.149.128 port 42300 ssh2
2026-02-21T09:38:23-05:00 kali sshd-session[7215]: Connection closed by invalid user brute 172.16.149.128 port 42300 [preauth]
2026-02-21T09:38:23-05:00 kali sshd-session[7215]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=172.16.149.128
2026-02-21T09:38:23-05:00 kali sshd-session[7345]: Invalid user brute from 172.16.149.128 port 60762
```

Final Check

- Check Fail2Ban afterwards:

```
sudo fail2ban-client status sshd
```

```
┌──(kali㉿kali)-[~]
└─$ sudo fail2ban-client status sshd
Status for the jail: sshd
├- Filter
|  ├- Currently failed: 0
|  ├- Total failed:     25
|  `- Journal matches:  _SYSTEMD_UNIT=ssh.service + _COMM=sshd
`- Actions
   ├- Currently banned: 1
   ├- Total banned:     2
   `- Banned IP list:   172.16.149.128
```

Deliverables

- Screenshots of logs before and after enabling Fail2Ban
- Screenshots of fail2ban status before and after brute ssh attempts
- Bonus (optional) how would you keep attempting ssh connections after being blacklisted by Fail2Ban

Thank You!